1️⃣ Manipulate numbers, graphical and statistical data (Item 7)

Relevance: ★★★★★ (Most critical)

Description:

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

Manifestation in this problem:

Students must perform several numerical operations:

- Subtraction (removing the membership fee)
- Division (calculating the number of DVDs rented)
- Multiplication (calculating the non-member total cost)

The calculations include:

$$52.50 - 10$$

$$42.50 \div 2.50$$

$$17 \times 3.20$$

Therefore, **numerical manipulation is the core competency required for this task**.

2️⃣ Use a multi-step procedure leading to a mathematical solution (Item 12)

Relevance: ★★★★★

Description:

> Use a multi-step procedure leading to a mathematical solution.

The problem requires several steps:

1. Understand the membership fee structure
2. Subtract the membership fee from the total cost
3. Calculate the number of DVDs rented
4. Recalculate the total cost using the non-member price

Thus, this task represents a **typical multi-step problem-solving process**.

3️⃣ Make sense of, relate and use a variety of representations (Item 10)

Relevance: ★★★★☆

Description:

> Make sense of, relate and use a variety of representations when interacting with a problem.

The problem presents information in multiple forms:

- A contextual description
- A price table

Students must translate between representations:

Table information → mathematical calculations

For example:

- Member price
- Non-member price

4️⃣ Understand and utilise constructs based on definitions and rules (Item 5)

Relevance: ★★★★☆

Description:

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

The problem provides clear pricing rules:

- Member rental fee
- Non-member rental fee
- Annual membership fee

Students must understand these rules and apply them correctly.

5️⃣ Use an understanding of the context to guide solving (Item 13)

Relevance: ★★★☆☆

Description:

> Use an understanding of the context to guide solving.

Students must understand the real-world context:

- DVD rental pricing
- Membership systems
- Different pricing structures

This contextual understanding helps construct the mathematical model.

6️⃣ Articulate a solution (Item 8)

Relevance: ★★★☆☆

Description:

> Articulate a solution, showing and/or summarising intermediate mathematical results.

Although the problem only asks for the final answer, a complete solution typically includes:

- The number of DVDs rented
- The calculation of the non-member cost

Thus, it involves some level of **explanation and presentation of reasoning**.

1️⃣ Manipulate numbers, graphical and statistical data（第7项）

相关性：★★★★★（最核心）

能力描述：

> Manipulate numbers, graphical and statistical data and information, algebraic expressions anc equations.

在本题中的体现：

学生需要进行多种数值运算：

- 减法（去掉会员费）
- 除法（计算 DVD 数量）
- 乘法（计算非会员费用）

涉及的计算包括：

$$52.50 - 10$$

$$42.50 \div 2.50$$

$$17 \times 3.20$$

因此 **数值操作能力是本题最核心能力。**

2️⃣ Use a multi-step procedure leading to a mathematical solution（第12项）

相关性：★★★★★

能力描述：

> Use a multi-step procedure leading to a mathematical solution.

解题过程需要多个步骤：

1. 理解会员费用结构
2. 从总费用中减去会员费
3. 计算 DVD 数量
4. 使用非会员价格重新计算总费用

因此属于 **典型多步骤问题求解任务。**

3️⃣ Make sense of, relate and use a variety of representations（第10项）

相关性：★★★★☆

能力描述：

> Make sense of, relate and use a variety of representations when interacting with a problem.

题目提供了多种信息形式：

- 文字情境描述
- 价格表

学生需要进行转换：

表格信息 → 数学计算

例如：

- 会员价格
- 非会员价格

4️⃣ Understand and utilise constructs based on definitions and rules（第5项）

相关性：★★★★☆

能力描述：

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

题目包含明确的收费规则：

- 会员价格
- 非会员价格
- 年费

学生需要理解这些规则，并正确应用。

5️⃣ Use an understanding of the context to guide solving（第13项）

相关性：★★★☆☆

能力描述：

> Use an understanding of the context to guide solving.

学生需要理解现实情境：

- DVD 租赁收费
- 会员制度
- 不同价格结构

理解这些背景有助于建立数学模型。

6️⃣ Articulate a solution（第8项）

相关性：★★★☆☆

能力描述：

> Articulate a solution, showing and/or summarising intermediate mathematical results.

虽然题目只要求最终答案，但完整解题过程需要展示：

- DVD 数量
- 非会员费用计算

因此涉及一定程度的 **表达和解释能力。**